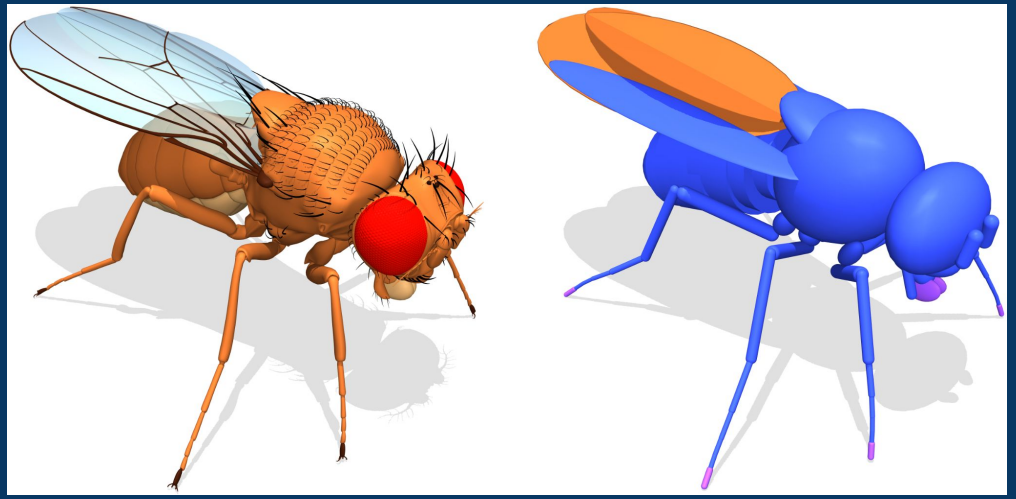
Evolving a Fruit Fly Flight Controller Using Neuroevolution of Augmenting Topologies

Matthew Lee

Prof.Gary Parker, Jim O'Connor

Drosophila Melanogaster



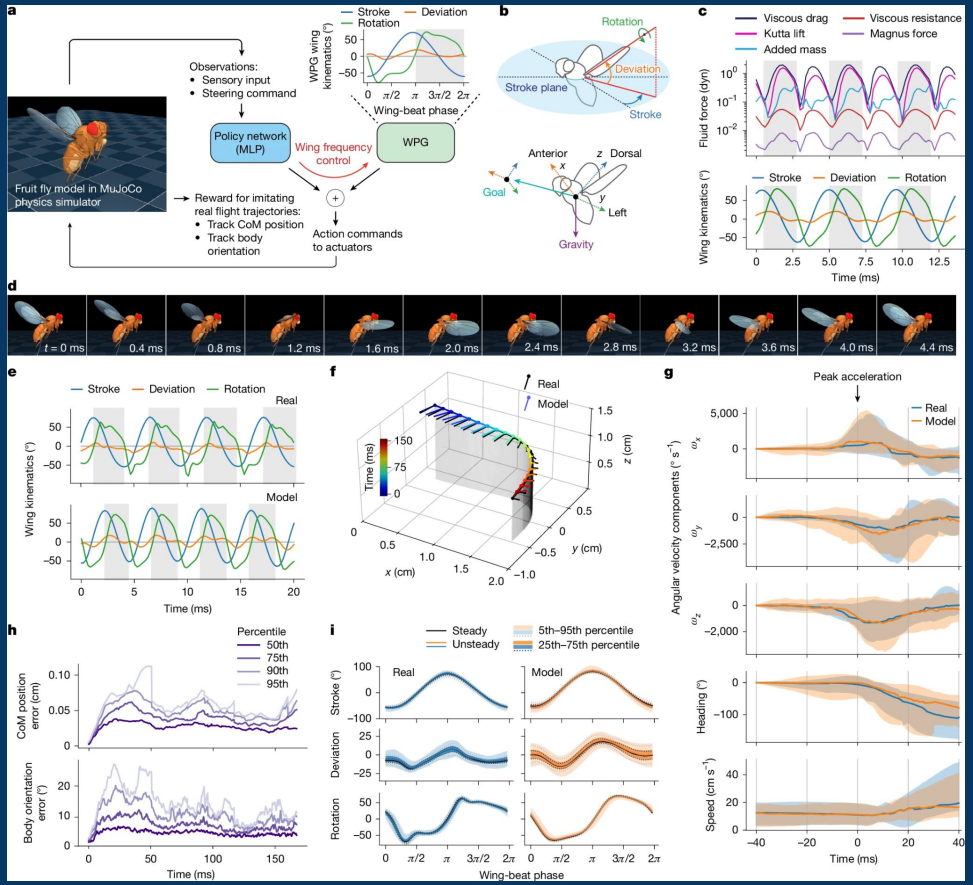


- Model Organism as proposed by Charles W. Woodworth in 1901
- Widely used species for studying in various fields
- Anatomically accurate model recreated in Mujoco physics simulation

Related Work

- Whole-body physics simulation of fruit fly locomotion
  - A fruit fly controller trained for walking, flying, and navigating using reinforcement learning

Roman Vaxenburg, Igor Siwanowicz, Josh Merel, Alice A Robie, Carmen Morrow, Guido Novati, Zinovia Stefanidi, Gert-Jan Both, Gwyneth M Card, Michael B Reiser, Matthew M Botvinick, Kristin M Branson, Yuval Tassa, & Srinivas C Turaga (2025). Whole-body physics simulation of fruit fly locomotion. *Nature*.



Concepts

Genetic Algorithm

Neuroevolution of Augmenting Topology (NEAT)

Punctuated Anytime Learning (PAL)

Neuro-PAL

Recurrent Neural Network

Genetic Algorithm Overview

- An optimization algorithm inspired by evolution
- The optimized solutions are chromosomes
- Fitness evaluation of chromosomes
- Selection and crossover between genes
- Random mutation of chromosomes
- Survival of the fittest (Extinction of weak species)

Concepts

Genetic Algorithm

Neuroevolution of Augmenting Topology (NEAT)

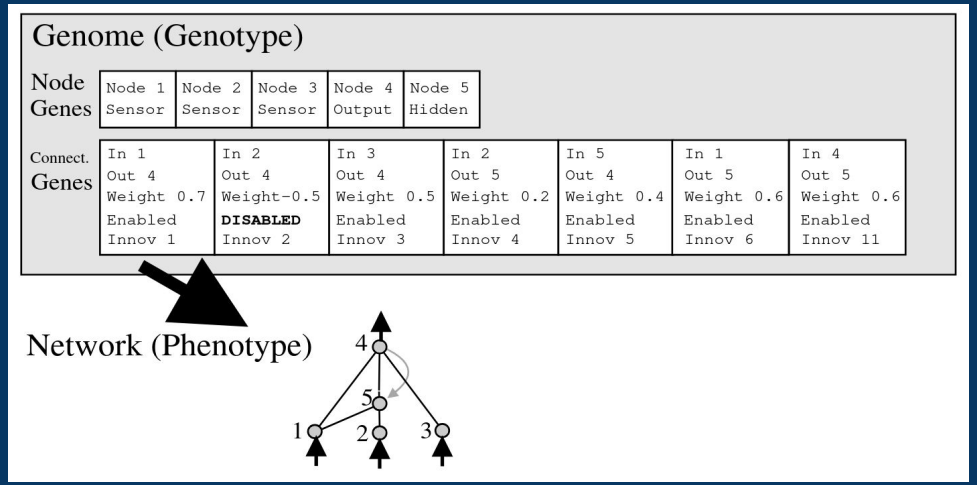
Punctuated Anytime Learning (PAL)

Neuro-PAL

Recurrent Neural Network

Neuroevolution of Augmenting Topology (NEAT)

- Each chromosome represents a neural network
- Optimizes its weights and the architecture
- Creation and deletion of nodes and connections through evolutionary process
- Can optimize a neural network with simplified structure



Stanley, Kenneth O., and Risto Miikkulainen. "Evolving neural networks through augmenting topologies." *Evolutionary computation* 10.2 (2002): 99-127.

Concepts

Genetic Algorithm

Neuroevolution of Augmenting Topology (NEAT)

Punctuated Anytime Learning (PAL)

Neuro-PAL

Recurrent Neural Network

Punctuated Anytime Learning

- Originally developed as a real-time adaptation method
- Combines two different evaluation functions to improve learning speed (Accurate/slow and loose/fast)
- Accelerates learning
  - Coevolution of model parameters
  - Fitness-biasing
- In fitness-biasing method, we bias the selection process towards chromosomes with higher fitness

Concepts

Genetic Algorithm

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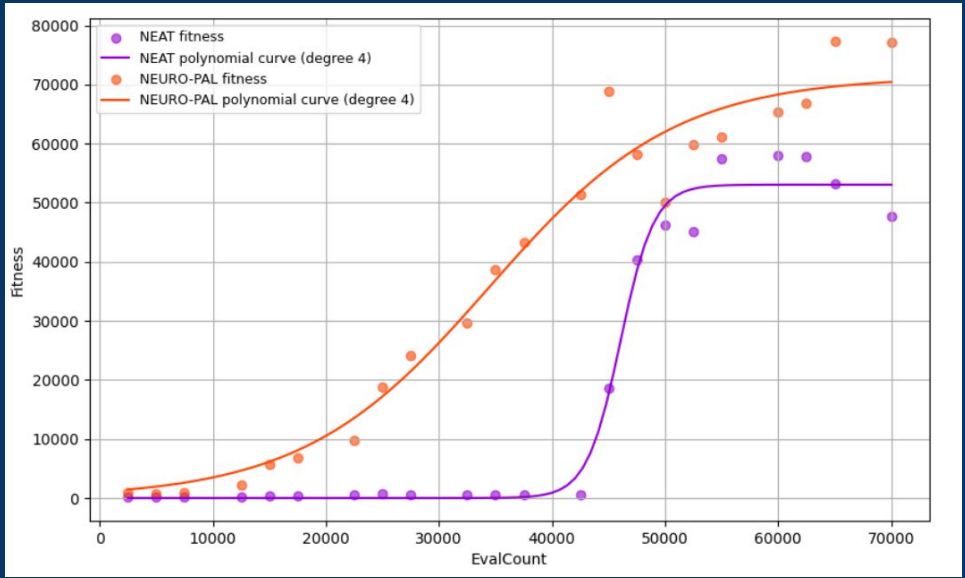
Punctuated Anytime Learning (PAL)

Neuro-PAL

Recurrent Neural Network

Neuro-PAL

- Combines NEAT with PAL
- Fitness-biasing
- Two evaluation functions with different evaluation count
- Shown to be an effective at increasing fitness



O'Connor, Jim, Yeonghun Lee, and Gary B. Parker. "NeuroPAL: Punctuated Anytime Learning with Neuroevolution for Macromanagement in Starcraft: Brood War." *arXiv preprint arXiv:2506.10384* (2025).

Concepts

Genetic Algorithm

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Neuro-PAL

Recurrent Neural Network

Recurrent Neural Network

- Introduction of hidden state
- Memory across multiple steps
- Used in various time-series data processing neural networks such as flight controlling neural networks
- Recurrent node can be added to NEAT network

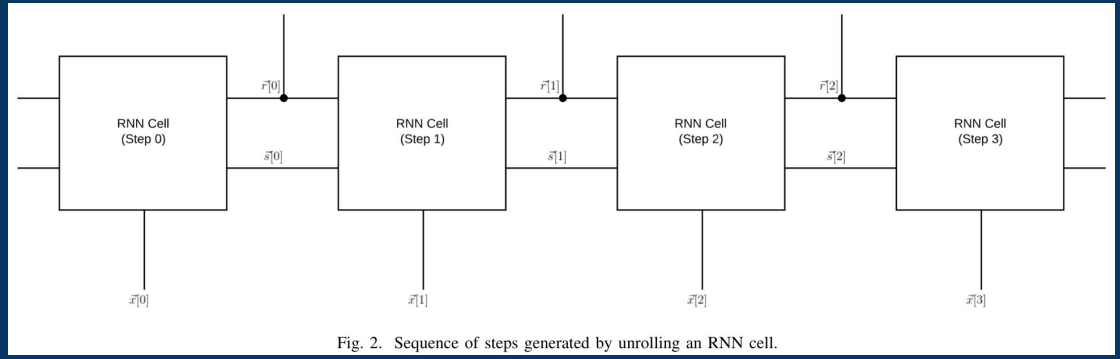
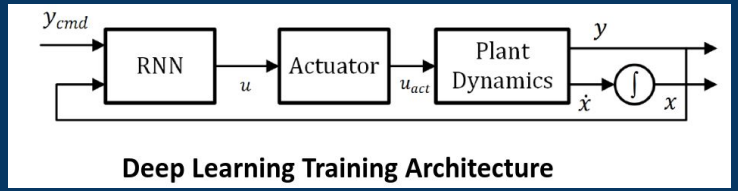


Fig. 2. Sequence of steps generated by unrolling an RNN cell.

Sherstinsky, Alex. "Fundamentals of recurrent neural network (RNN) and long short-term memory (LSTM) network." *Physica D: Nonlinear Phenomena* 404 (2020): 132306.



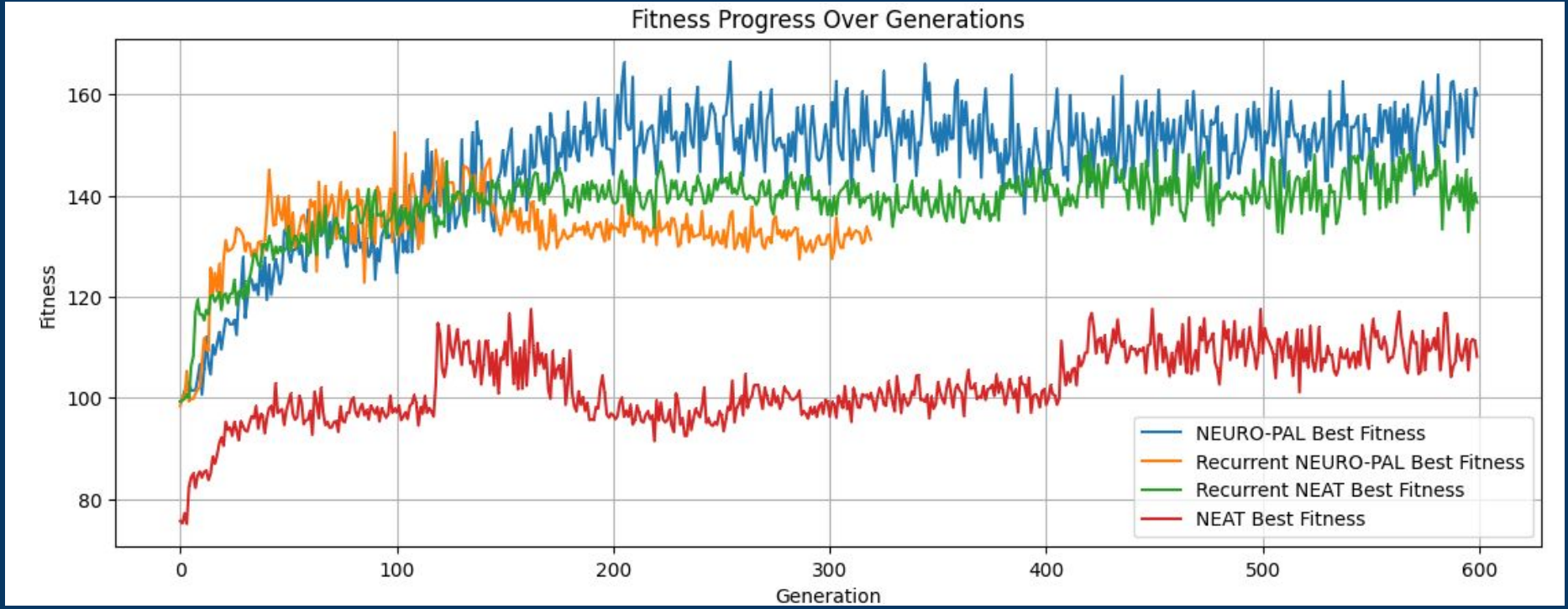
Deep Learning Training Architecture

Nivison, Scott A., and Pramod P. Khargonekar. "Development of a robust deep recurrent neural network controller for flight applications." *2017 American Control Conference (ACC)*. IEEE, 2017.

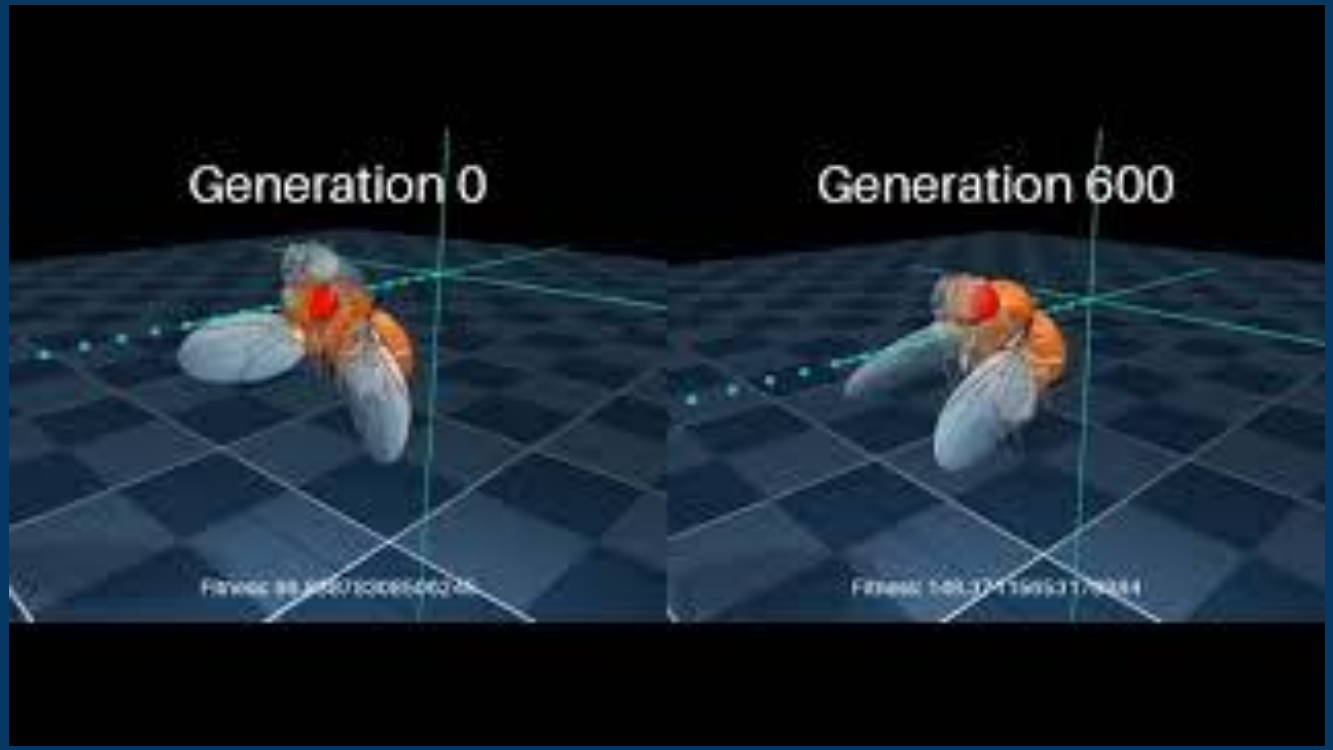
Methodology

- NEAT
- Recurrent NEAT
- Neuro-PAL (NEAT PAL)
- Recurrent Neuro-PAL

Initial Results



Video Demonstration



Thank You